

Diabetes: A Comprehensive Medical Guide

1. Overview

Diabetes mellitus is a chronic metabolic disease characterized by elevated levels of blood glucose, which leads over time to serious damage to the heart, blood vessels, eyes, kidneys, and nerves. There are three main types: Type 1, Type 2, and Gestational diabetes. Globally, an estimated 422 million people have diabetes, and 1.5 million deaths are directly attributed to the disease each year. The number of cases has been steadily rising over the past few decades, particularly in low- and middle-income countries.

2. Types of Diabetes

Type 1 Diabetes: An autoimmune condition in which the body's immune system attacks and destroys insulin-producing beta cells in the pancreas. It requires daily insulin administration and typically develops in children and young adults.

Type 2 Diabetes: The most common form, accounting for around 90% of all diabetes cases. It occurs when the body does not use insulin effectively. It is strongly linked to obesity, physical inactivity, and poor diet. It can often be managed through lifestyle changes.

Gestational Diabetes: A form of diabetes that develops during pregnancy and usually resolves after delivery. However, it increases the risk of developing Type 2 diabetes later in life.

3. Symptoms

Common symptoms of diabetes include:

- Frequent urination (polyuria)
- Excessive thirst (polydipsia)
- Unexplained weight loss
- Extreme fatigue and lack of energy
- Blurred vision
- Slow healing of wounds and cuts
- Tingling, numbness, or pain in hands and feet (neuropathy)
- Recurrent infections (skin, gum, or bladder infections)

Type 1 symptoms often appear suddenly, while Type 2 symptoms may develop gradually and go unnoticed for years.

4. Risk Factors

The following factors increase the risk of developing Type 2 diabetes:

- Being overweight or obese (BMI over 25)

- Physical inactivity and sedentary lifestyle
- Family history of diabetes
- Age over 45 years
- High blood pressure (hypertension)
- History of gestational diabetes
- Polycystic ovary syndrome (PCOS)
- Unhealthy diet high in sugar and processed foods
- Smoking and excessive alcohol consumption
- Ethnicity (higher risk in African, Hispanic, Asian, and Indigenous populations)

5. Diagnosis

Diabetes is diagnosed through blood tests, including:

- Fasting Blood Glucose Test: A reading of 126 mg/dL or higher after an 8-hour fast indicates diabetes.
- HbA1c Test (Glycated Haemoglobin): An HbA1c level of 6.5% or above confirms diabetes.
- Oral Glucose Tolerance Test (OGTT): A 2-hour blood glucose of 200 mg/dL or higher indicates diabetes.
- Random Blood Sugar Test: A reading of 200 mg/dL or higher along with symptoms confirms diagnosis.

Pre-diabetes is diagnosed when blood glucose levels are higher than normal but not yet in the diabetic range. Early detection is critical to preventing progression.

6. Treatment and Management

Treatment depends on the type and severity of diabetes:

Medication: Type 1 diabetes requires insulin therapy. Type 2 diabetes may be managed with oral medications such as Metformin, SGLT2 inhibitors, or GLP-1 receptor agonists. Some Type 2 patients also require insulin.

Blood Sugar Monitoring: Regular self-monitoring using a glucometer helps patients track their glucose levels and adjust treatment accordingly.

Diet: A balanced diet low in refined sugars, saturated fats, and processed foods is essential. Emphasis should be placed on whole grains, vegetables, lean proteins, and healthy fats.

Exercise: At least 150 minutes of moderate-intensity aerobic activity per week is recommended. Physical activity improves insulin sensitivity and helps maintain healthy body weight.

Regular Check-ups: Patients should have regular screenings for complications including eye exams, kidney function tests, blood pressure monitoring, and foot examinations.

7. Complications

Uncontrolled diabetes can lead to serious complications:

- Cardiovascular disease: Heart attacks and strokes are 2-3 times more common in diabetic patients.
- Diabetic nephropathy: Kidney damage leading to chronic kidney disease or kidney failure.
- Diabetic retinopathy: Damage to blood vessels in the retina, a leading cause of blindness.
- Peripheral neuropathy: Nerve damage causing pain, numbness, and weakness in limbs.
- Diabetic foot: Poor circulation and neuropathy can lead to foot ulcers and, in severe cases, amputation.
- Increased susceptibility to infections due to weakened immune function.

8. Prevention

Type 2 diabetes can largely be prevented through lifestyle modifications:

- Maintain a healthy body weight through regular physical activity
- Follow a nutritious, balanced diet rich in fibre and low in sugar
- Avoid smoking and limit alcohol consumption
- Get regular health screenings, especially if you have risk factors
- Manage blood pressure and cholesterol levels
- Breastfeed infants when possible, as it reduces diabetes risk in both mother and child

Public health interventions, community awareness, and early screening programs play a critical role in reducing the global burden of diabetes.

9. Global Statistics

- 422 million people worldwide live with diabetes (WHO, 2023)
- 1 in 10 adults globally has diabetes
- 1.5 million deaths are directly caused by diabetes annually
- Over 80% of diabetes deaths occur in low- and middle-income countries
- Type 2 diabetes accounts for approximately 90-95% of all diabetes cases
- Africa has the highest proportion of undiagnosed diabetes cases globally
- The global cost of diabetes-related healthcare exceeds \$760 billion annually

10. When to See a Doctor

You should consult a healthcare professional if you experience:

- Persistent thirst or frequent urination
- Unexplained weight loss over a short period
- Fatigue that does not improve with rest
- Blurred vision or sudden changes in eyesight
- Slow-healing wounds or recurring infections
- Numbness or tingling in your hands or feet

Early diagnosis and prompt treatment significantly reduce the risk of complications and improve long-term quality of life.